

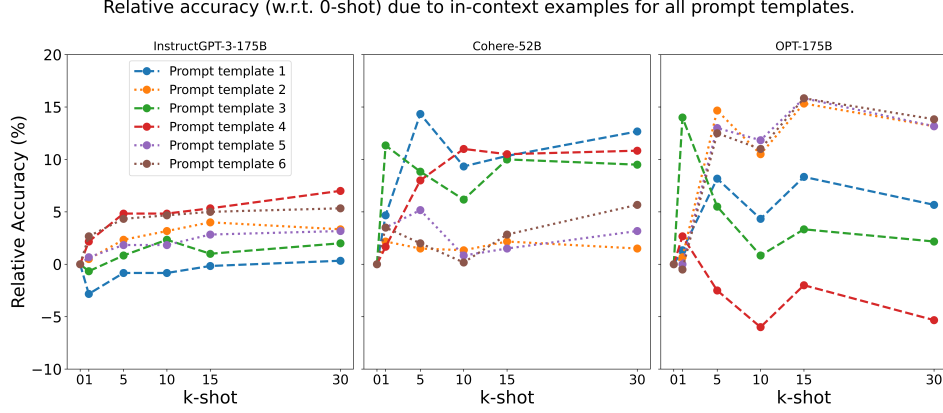


Figure 7: Relative performance increase over 0-shot due to in-context prompting. Structured prompt templates are dashed lines (1, 3, 4) and natural prompt templates dotted lines (2, 5, 6).

K.5 Variance over API runs

In this section we comment on the reproducibility of research done using APIs. OpenAI and Cohere have their models behind an API, meaning we do not have control over what happens to the prompt before the model processes it. We run the zero-shot evaluation ten more times for two models of OpenAI and Cohere, text-davinci-002 and Cohere-52B. The results from this experiment are shown in Table 14 and 15. From this we can conclude that there is some stochasticity in the API that we have no control over, a bit more for OpenAI than for Cohere, but again we can be relatively confident that the conclusion will not be different because of it. The results from this work are therefore reproducible with access to the same models behind the API now. Unfortunately, when OpenAI or Cohere changes the models behind the API, these results are not exactly reproducible anymore.

For completeness, we add the timestamp that each result was obtained below (Appendix L).

Table 14: Results per prompt template (P.T.) for 10 different runs from text-davinci-002 for 0-shot evaluation.

Each evaluation has exactly the same text, so the variance in performance is due to API stochasticity.

API-run	P. T. 1	P. T. 2	P. T. 3	P. T. 4	P. T. 5	P. T. 6	Mean
0	73.50	68.83	73.00	71.17	67.17	68.83	70.42
1	73.83	69.00	72.83	71.50	67.67	68.33	70.53
2	73.67	68.67	73.17	71.33	67.50	68.50	70.47
3	73.83	68.17	73.17	71.00	67.67	68.17	70.33
4	73.67	68.83	73.33	71.17	67.00	68.33	70.39
5	73.83	68.50	73.00	71.00	67.00	68.17	70.25
6	73.67	69.00	73.00	71.17	67.33	68.50	70.44
7	73.67	68.67	72.83	71.33	67.50	68.67	70.44
8	73.83	69.17	72.83	71.17	67.33	68.00	70.39
9	73.50	68.50	72.83	71.00	67.50	68.67	70.33
10	73.67	69.50	73.00	71.33	67.50	68.50	70.58
mean	73.70	68.80	73.00	71.20	67.38	68.42	-
std	0.12	0.35	0.16	0.16	0.23	0.24	-

K.6 Experiment with random in-context labels

This paper presents the thesis that instruction-tuning at the example level (“Example IT”) is important for pragmatic understanding in LLMs. However, the 0-shot result that one of the models in the Example IT group achieves is similar to that of base models; Cohere-command-52b obtains a zero-shot performance of 60.2%. From the sharp rise in performance observed for the $k = 0$ to $k = 1$ result (from 60.2% to 72.8%) we hypothesise that the k-shot in-context examples in this task do not necessarily teach the model pragmatics in-context, but prime the model for the task format (namely,

Table 15: Results per prompt template (P.T.) for 10 different runs from Cohere-52B for 0-shot evaluation.

Each evaluation has exactly the same text, so the variance in performance is due to API stochasticity.

API-run	P. T. 1	P. T. 2	P. T. 3	P. T. 4	P. T. 5	P. T. 6	Mean
0	56.00	62.67	54.33	54.00	62.17	62.17	58.56
1	56.00	62.83	54.33	54.00	62.33	62.33	58.64
2	56.00	62.83	54.33	54.00	62.17	62.33	58.61
3	56.00	62.83	54.33	54.00	62.17	62.33	58.61
4	55.83	62.67	54.33	54.00	62.17	62.33	58.56
5	56.00	62.83	54.33	54.00	62.17	62.17	58.58
6	56.00	62.83	54.33	54.00	62.17	62.17	58.58
7	56.00	62.67	54.33	54.00	62.33	62.17	58.58
8	56.00	62.83	54.33	54.00	62.00	62.33	58.58
9	56.00	62.83	54.00	53.83	62.17	62.17	58.50
mean	55.98	62.78	54.30	53.98	62.18	62.25	-
std	0.05	0.08	0.10	0.05	0.09	0.08	-

outputting either “yes” or “no” as detailed in Section 3 in the main text). If this hypothesis is true, we would observe similar performance regardless of whether the labels given in the prompt for the few-shot examples are true. We test this empirically for two base models (GPT-3, Cohere-52b) and two Example IT models (text-davinci-001, Cohere-command-52b) for 1-shot and 5-shot evaluation. The results can be found in Table 16. We find that for the Example IT models in-context prompts with random labels obtain the same results (i.e. within confidence intervals) as the experiments with ground-truth labels in the in-context examples. For base models however we do observe a drop in performance; for GPT-3-175b at 5-shot, and Cohere-52b both at 1- and 5-shot. Taken together, we can conclude that for base models the content of the in-context prompt seems important, whereas for models in the example IT group the in-context examples mainly serve as a primer for the task structure.

Table 16: The results of the 1- and 5-shot experiment with random labels for the few-shot examples as opposed to the the true labels. We find that performance does not degrade for the models in the Example IT group, which implies that for these models not the content of the examples is important for performance, but the structure.

Model	1-shot	1-shot rand labels	5-shot	5-shot rand labels
GPT-3-175b	65.7% $\pm$ 1.4	65.4% $\pm$ 1.2	68.7% $\pm$ 1.5	64.7% $\pm$ 1.9
Cohere-52b	63.0% $\pm$ 3.8	58.3% $\pm$ 3.3	65.1% $\pm$ 2.9	60.5% $\pm$ 1.9
text-davinci-001	72.7% $\pm$ 1.3	73.9% $\pm$ 1.7	74.5% $\pm$ 1.0	73.4% $\pm$ 1.2
Cohere-command-52b	72.8% $\pm$ 1.3	72.0% $\pm$ 1.6	75.4% $\pm$ 1.8	73.5% $\pm$ 2.7

Table 17: Results of the chain-of-thought (CoT) experiment for models in the base group. The numbers between brackets show the difference in performance with the number on the same row one column to the left. These models do not benefit from CoT-prompting. The reason Cohere-6b achieves such a low score for CoT-prompting is because it is not able to adhere to the correct output format (yes/no).

Model	0-shot	5-shot	5-shot CoT
GPT-3-350m	51.5% $\pm$ 3.0	55.7% $\pm$ 1.6 (+4.2%)	55.0% $\pm$ 3.5 (-0.7%)
GPT-3-1.3b	57.7% $\pm$ 3.1	62.6% $\pm$ 2.0 (+4.9%)	54.4% $\pm$ 5.8 (-8.2%)
GPT-3-6.7b	54.8% $\pm$ 1.9	62.4% $\pm$ 1.5 (+7.6%)	61.0% $\pm$ 2.3 (+4.0%)
GPT-3-175b	57.2% $\pm$ 4.4	68.7% $\pm$ 1.5 (+11.5%)	60.3% $\pm$ 4.2 (-8.4%)
Cohere-6b	57.3% $\pm$ 2.2	60.9% $\pm$ 4.1 (+3.6%)	29.2% $\pm$ 14.7 (-31.7%)
Cohere-52b	58.5% $\pm$ 4.0	65.1% $\pm$ 2.9 (+6.6%)	64.7% $\pm$ 3.2 (-0.4%)

K.7 Chain-of-thought on base models

In the main paper we do a CoT experiment on the models in the Example IT group. Base models also benefit from in-context examples, so it makes sense to also try CoT prompting on these models. After attempting this for two of the model classes in the group, we decided not to apply this prompting technique to the other models, because it decreases performance, sometimes significantly. See the results of the CoT experiment on the two base model classes in Table 17

K.8 Testing for spurious correlations

In this section, we do a small scale experiment to test whether the benchmark has spurious correlations. Specifically, we run the benchmark with only the utterance or only the response as input. Strictly, getting the implicature right from the response only does not always indicate spurious correlations, as some examples only need the response (e.g. rhetorical questions like ‘do pigs fly?’). Utterance-only results do always indicate spurious correlations. We run this experiment for GPT-3.5-turbo and GPT-4 0-shot and 5-shot (see Table 18 and Table 19).

Table 18: Results of running the benchmark with only the utterance as input, to test for spurious correlations with the label.

Utterance-only	0-shot	5-shot
GPT-3.5-Turbo	54.3% $\pm$ 3.3	41.7% $\pm$ 12.4
GPT-4	48.9% $\pm$ 10.5	53.7% $\pm$ 0.5

Table 19: Results of running the benchmark with only the response as input, to test what part of the examples can be resolved without the utterance.

Response-only	0-shot	5-shot
GPT-3.5-Turbo	59.2% $\pm$ 4.7	58.3% $\pm$ 6.6
GPT-4	62.6% $\pm$ 1.7	65.5% $\pm$ 1.1

We find that models mostly perform random for utterance-only, so spurious correlations do not seem to be an issue. For response-only, GPT-4 5-shot gets 65% accuracy. Some examples it gets right are: “do fish swim?” and “let’s hope so”.